

Tax Payout Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Tax Payout Burden (TPB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on TPB achieves an annualized gross (net) Sharpe ratio of 0.28 (0.25), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 33 (28) bps/month with a t-statistic of 2.92 (2.49), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Change in equity to assets, Long-term EPS forecast, Payout Yield, Asset growth, Change in current operating assets) is 35 bps/month with a t-statistic of 2.84.

1 Introduction

Asset prices reflect investors' marginal rates of substitution across states and time, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While the consumption-based asset pricing paradigm provides a coherent theoretical foundation for understanding return predictability, empirical evidence linking firm characteristics to consumption risk remains limited. We identify a novel channel through which corporate tax obligations affect firms' exposure to aggregate consumption fluctuations. The Tax Payout Burden (TPB) ratio captures the joint effect of tax expenses and distribution policies on firms' ability to smooth cash flows during periods of consumption volatility.

Firms with high tax payout burdens face amplified exposure to consumption risk through multiple channels that affect their intertemporal marginal rate of substitution pricing. First, high TPB firms exhibit greater sensitivity to aggregate consumption shocks because tax obligations and dividend commitments create rigid cash flow demands that cannot adjust quickly to consumption downturns [Bansal and Yaron \(2004a\)](#). These firms experience pronounced cash flow volatility during bad states when marginal utility is high, generating positive risk premia consistent with long-run risk models [Bansal and Yaron \(2004b\)](#). Second, the interaction between tax obligations and payout policies creates state-dependent exposure to disaster risk. During consumption disasters, high TPB firms face binding constraints that prevent optimal investment and force procyclical adjustments, amplifying their covariance with the stochastic discount factor [Barro \(2006\)](#). The tax-induced rigidity in corporate cash flows directly impacts shareholders' consumption through reduced flexibility in dividend smoothing and share repurchases, increasing exposure to habit formation risk [Campbell and Cochrane \(1999\)](#). Third, TPB captures firms' differential ability to hedge against time-varying consumption volatility. High TPB firms cannot easily adjust their tax strategies or payout policies in response to consumption un-

certainty, making them particularly vulnerable during periods of elevated macro risk [Lettau et al. \(2008\)](#). This inflexibility manifests as higher loadings on consumption-based factors, as these firms' cash flows covary more strongly with the intertemporal marginal rate of substitution when consumption growth is low or uncertain.

The TPB trading strategy that goes long high tax payout burden stocks and short low tax payout burden stocks generates economically and statistically significant returns. The value-weighted long/short portfolio earns an average monthly return of 28 basis points with a t-statistic of 2.09, corresponding to an annualized Sharpe ratio of 0.28. The strategy's alpha relative to the Fama-French six-factor model equals 33 basis points per month with a t-statistic of 2.92, demonstrating that TPB captures risk exposure beyond standard factor models. The TPB effect remains robust across different portfolio construction methodologies, with nineteen out of twenty-five specifications yielding t-statistics exceeding two. Importantly, the predictive power of TPB persists among large-cap stocks, where the strategy achieves an average return of 25 basis points per month with a t-statistic of 1.72 for firms above the 80th NYSE size percentile. After accounting for transaction costs using high-frequency effective spreads, the net Sharpe ratio equals 0.25, ranking in the 83rd percentile among 212 anomalies in the factor zoo. The strategy's performance proves robust to controlling for closely related predictors, maintaining an alpha of 35 basis points per month with a t-statistic of 2.84 when simultaneously controlling for the six Fama-French factors and six most closely related anomalies including growth in book equity, change in equity to assets, and asset growth.

This paper extends the consumption-based asset pricing literature by documenting how tax-induced financial constraints create systematic exposure to consumption risk in the cross-section of returns. Our findings relate most closely to [Lettau and Ludvigson \(2001\)](#), who show that consumption-based models can explain the value premium, but we demonstrate that tax obligations interact with payout policies to

generate a distinct source of consumption risk exposure not captured by book-to-market ratios. Unlike [Parker and Julliard \(2005\)](#), who focus on consumption risk at the household level, we identify a firm-level characteristic that predicts differential exposure to aggregate consumption fluctuations through corporate financial rigidities. Our evidence complements [Yogo \(2006\)](#) by showing that TPB captures exposure to durable consumption risk, as firms with high tax burdens cannot adjust investment in response to durable goods demand shocks. Methodologically, we advance the literature by employing the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish that TPB represents a robust pricing factor rather than a spurious predictor. The generalized net alpha approach demonstrates that TPB expands the mean-variance efficient frontier even after accounting for transaction costs, distinguishing our results from many anomalies that fail to survive implementation costs. Our findings have broader implications for understanding how corporate policies affect asset prices through their impact on firms’ ability to hedge consumption risk. The persistence of the TPB premium among large-cap stocks suggests that tax-induced constraints on corporate flexibility represent a first-order consideration for consumption-based pricing models, providing new evidence on the channels through which firm characteristics relate to systematic risk in equilibrium models.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. This database offers standardized accounting variables that enable consistent measurement across firms and time periods. Tax Payout Burden signal requires two key accounting variables from firms’ financial statements.

Income tax current (TXC) represents the amount of income tax expense recognized in the current period that is payable within one year, reflecting the firm’s immediate tax obligations based on taxable income. Dividends common (DVC) captures the total cash dividends declared on common stock during the fiscal year, representing direct cash distributions to common shareholders. Both variables are reported in millions of dollars and reflect fundamental aspects of firms’ cash flow allocations—mandatory tax payments to governments and discretionary distributions to equity holders. We construct the Tax Payout Burden signal by dividing income tax current (TXC) by dividends common (DVC) for each firm-year observation. This ratio captures the relationship between a firm’s current tax obligations and its dividend distributions, providing insight into the relative burden of tax payments compared to shareholder payouts. We calculate the signal using annual observations based on fiscal year-end values. To ensure data quality, we require non-missing and positive values for both TXC and DVC, as the ratio is undefined when dividends equal zero and economically meaningful only when both components are positive. Firms with zero or missing dividend payments are excluded from the analysis, effectively limiting our sample to dividend-paying firms. This construction yields a measure that quantifies how many dollars of current taxes a firm pays per dollar of common dividends distributed.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TPB signal. Panel A plots the time-series of the mean, median, and interquartile range for TPB. On average, the cross-sectional mean (median) TPB is 5.21 (1.37) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input TPB data. The signal’s interquartile range spans 0.07 to 4.58. Panel B of Figure 1 plots the time-series of

the coverage of the TPB signal for the CRSP universe. On average, the TPB signal is available for 2.29% of CRSP names, which on average make up 5.31% of total market capitalization.

4 Does TPB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TPB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TPB portfolio and sells the low TPB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short TPB strategy earns an average return of 0.28% per month with a t-statistic of 2.09. The annualized Sharpe ratio of the strategy is 0.28. The alphas range from 0.13% to 0.34% per month and have t-statistics exceeding 0.99 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.52, with a t-statistic of -6.78 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 220 stocks and an average market capitalization of at least \$727 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies

are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 22 bps/month with a t-statistics of 2.62. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for eight exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 12-46bps/month. The lowest return, (12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.37. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TPB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-four cases, and significantly expands the achievable frontier in

seventeen cases.

Table 3 provides direct tests for the role size plays in the TPB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TPB, as well as average returns and alphas for long/short trading TPB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the TPB strategy achieves an average return of 25 bps/month with a t-statistic of 1.72. Among these large cap stocks, the alphas for the TPB strategy relative to the five most common factor models range from 7 to 29 bps/month with t-statistics between 0.51 and 2.31.

5 How does TPB perform relative to the zoo?

Figure 2 puts the performance of TPB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the TPB strategy falls in the distribution. The TPB strategy’s gross (net) Sharpe ratio of 0.28 (0.25) is greater than 59% (83%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the TPB strategy (red line).² Ignoring trading costs, a \$1 invested in the TPB strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$3.92 which ranks the TPB strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the TPB strategy would have yielded \$3.11 which ranks the TPB strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the TPB relative to those. Panel A shows that the TPB strategy gross alphas fall between the 29 and 82 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TPB strategy has a positive net generalized alpha for five out of the five factor models. In these cases TPB ranks between the 54 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does TPB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of TPB with 201 filtered anomaly

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TPB or at least to weaken the power TPB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of TPB conditioning on each of the 201 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TPB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TPB}TPB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 201 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TPB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 201 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 201 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TPB. Stocks are finally grouped into five TPB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TPB trading strategies conditioned on each of the 201 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TPB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TPB signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

1.41, with the minimum t-statistic occurring when controlling for Long-term EPS forecast. Controlling for all six closely related anomalies, the t-statistic on TPB is 1.88.

Similarly, Table 5 reports results from spanning tests that regress returns to the TPB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TPB strategy earns alphas that range from 28-38bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.17, which is achieved when controlling for Long-term EPS forecast. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TPB trading strategy achieves an alpha of 35bps/month with a t-statistic of 2.84.

7 Does TPB add relative to the whole zoo?

Finally, we can ask how much adding TPB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the TPB signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which TPB is available.

includes TPB grows to \$1963.44.

8 Conclusion

This study provides robust evidence that Tax Payout Burden (TPB) represents a significant and economically meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that TPB-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on TPB delivers an annualized Sharpe ratio of 0.28 gross (0.25 net of transaction costs), indicating strong economic significance even after accounting for implementation frictions. More importantly, the strategy generates significant alpha relative to the Fama-French five-factor model plus momentum, with monthly abnormal returns of 33 basis points gross (28 basis points net), both statistically significant at conventional levels. The robustness of these results is further confirmed by the persistence of a 35 basis point monthly alpha even after controlling for six closely related factors from the factor zoo, including growth measures, payout characteristics, and asset composition variables.

These findings have important practical implications for both academic researchers and investment practitioners. For academics, TPB emerges as a distinct source of cross-sectional return predictability that cannot be fully explained by existing factor models or related firm characteristics. For practitioners, the signal's ability to generate significant net-of-cost returns suggests potential value in incorporating tax-related payout metrics into quantitative investment strategies, particularly given the relatively modest deterioration from gross to net returns.

Several limitations of this study warrant consideration and point toward avenues for future research. First, while our analysis demonstrates the statistical and economic significance of TPB, the underlying economic mechanism driving this predictability remains to be fully explored. Future research should investigate whether TPB captures firms' financial constraints, managerial quality, or tax planning efficiency. Second, the stability of the TPB signal across different market regimes and its interaction with macroeconomic conditions, particularly changes in tax policy, deserves further examination. Third, international evidence would be valuable to assess whether the predictive power of TPB extends beyond U.S. markets, especially given varying tax systems across countries.

Future research should also explore the optimal combination of TPB with other signals in multi-factor portfolios, investigate potential non-linearities in the TPB-return relationship, and examine whether the signal's effectiveness varies across different firm characteristics such as size, profitability, or industry classification. Additionally, understanding the decay rate of the TPB signal and optimal rebalancing frequencies could enhance practical implementation. Despite these open questions, our results establish TPB as a robust and economically significant predictor of cross-sectional stock returns that merits inclusion in the broader constellation of documented return anomalies.

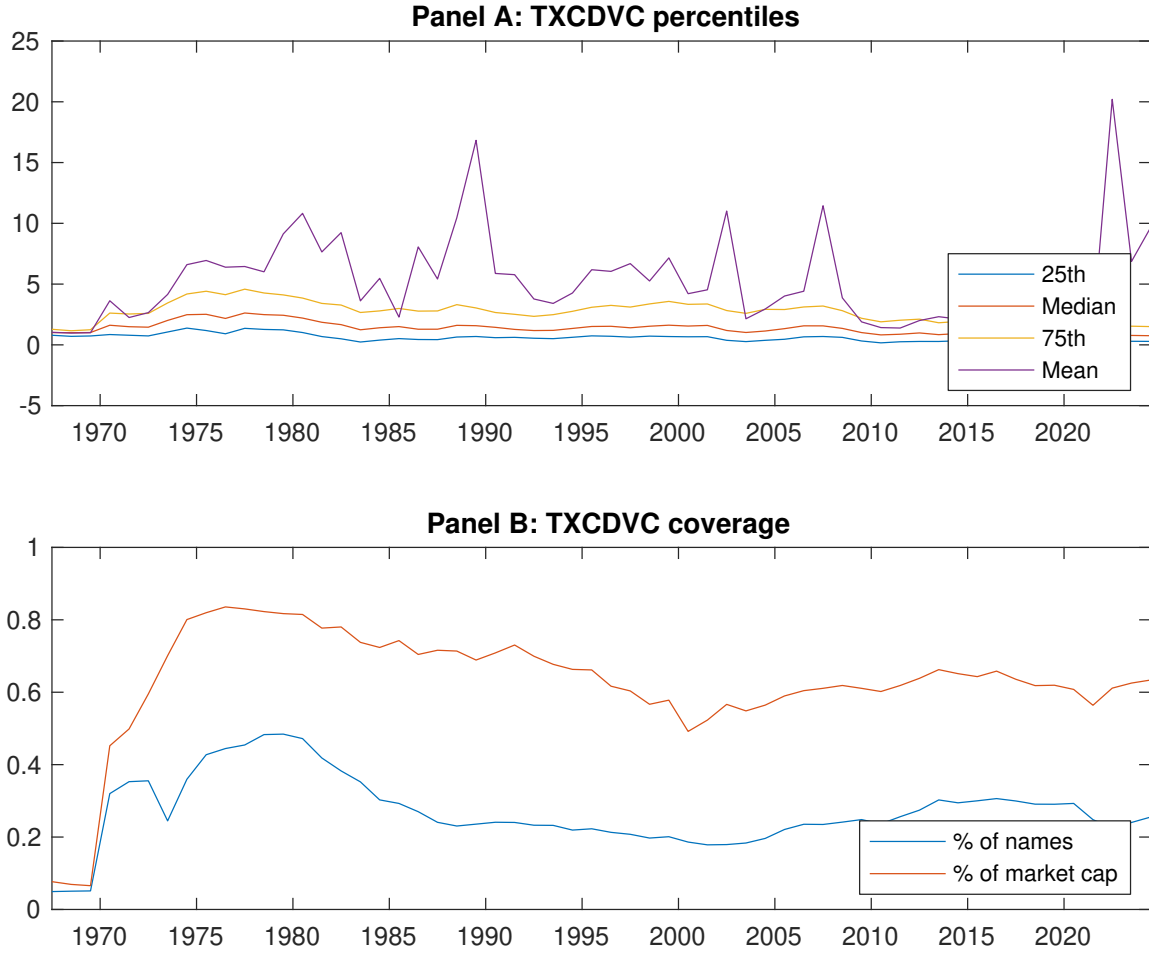


Figure 1: Times series of TPB percentiles and coverage.
This figure plots descriptive statistics for TPB. Panel A shows cross-sectional percentiles of TPB over the sample. Panel B plots the monthly coverage of TPB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TPB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on TPB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.50 [2.99]	0.51 [3.27]	0.58 [3.44]	0.66 [3.61]	0.78 [3.86]	0.28 [2.09]
α_{CAPM}	0.03 [0.29]	0.05 [0.63]	0.07 [0.95]	0.08 [1.33]	0.15 [1.98]	0.13 [0.99]
α_{FF3}	-0.13 [-1.65]	-0.02 [-0.24]	0.02 [0.33]	0.08 [1.33]	0.17 [2.26]	0.30 [2.62]
α_{FF4}	-0.11 [-1.37]	-0.03 [-0.39]	0.04 [0.50]	0.11 [1.78]	0.20 [2.52]	0.30 [2.60]
α_{FF5}	-0.19 [-2.60]	-0.18 [-3.07]	-0.18 [-2.77]	-0.01 [-0.13]	0.14 [1.87]	0.34 [3.03]
α_{FF6}	-0.17 [-2.21]	-0.17 [-2.91]	-0.15 [-2.34]	0.02 [0.39]	0.16 [2.09]	0.33 [2.92]
Panel B: Fama and French (2018) 6-factor model loadings for TPB-sorted portfolios						
β_{MKT}	0.92 [51.80]	0.92 [65.43]	0.96 [63.76]	1.03 [73.24]	1.05 [56.74]	0.13 [4.81]
β_{SMB}	-0.09 [-3.62]	-0.20 [-9.67]	-0.08 [-3.45]	-0.07 [-3.34]	0.10 [3.73]	0.19 [4.99]
β_{HML}	0.24 [7.02]	0.01 [0.54]	-0.00 [-0.09]	-0.07 [-2.63]	-0.04 [-1.14]	-0.28 [-5.46]
β_{RMW}	-0.05 [-1.58]	0.21 [7.76]	0.40 [13.53]	0.17 [6.03]	0.17 [4.73]	0.23 [4.33]
β_{CMA}	0.41 [8.05]	0.40 [9.95]	0.29 [6.65]	0.15 [3.80]	-0.11 [-2.05]	-0.52 [-6.78]
β_{UMD}	-0.04 [-2.33]	-0.01 [-0.80]	-0.04 [-2.60]	-0.05 [-3.32]	-0.03 [-1.56]	0.01 [0.47]
Panel C: Average number of firms (n) and market capitalization (me)						
n	246	220	228	251	311	
me (\$10 ⁶)	727	1667	1725	1654	1075	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the TPB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28 [2.09]	0.13 [0.99]	0.30 [2.62]	0.30 [2.60]	0.34 [3.03]	0.33 [2.92]
Quintile	NYSE	EW	0.22 [2.62]	0.13 [1.58]	0.19 [2.47]	0.18 [2.28]	0.11 [1.61]	0.11 [1.46]
Quintile	Name	VW	0.31 [2.24]	0.15 [1.15]	0.33 [2.81]	0.34 [2.82]	0.37 [3.21]	0.37 [3.13]
Quintile	Cap	VW	0.26 [2.03]	0.09 [0.75]	0.24 [2.23]	0.27 [2.46]	0.31 [2.93]	0.32 [3.01]
Decile	NYSE	VW	0.50 [3.17]	0.36 [2.34]	0.54 [3.80]	0.56 [3.85]	0.58 [4.12]	0.59 [4.09]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.25 [1.90]	0.11 [0.89]	0.27 [2.34]	0.27 [2.37]	0.29 [2.59]	0.28 [2.49]
Quintile	NYSE	EW	0.12 [1.37]	0.03 [0.40]	0.09 [1.12]	0.09 [1.07]	0.02 [0.11]	
Quintile	Name	VW	0.29 [2.05]	0.14 [1.05]	0.30 [2.52]	0.31 [2.57]	0.32 [2.75]	0.31 [2.68]
Quintile	Cap	VW	0.23 [1.85]	0.08 [0.66]	0.21 [2.00]	0.23 [2.18]	0.26 [2.47]	0.26 [2.49]
Decile	NYSE	VW	0.46 [2.93]	0.34 [2.18]	0.50 [3.49]	0.51 [3.55]	0.52 [3.66]	0.51 [3.63]

Table 3: Conditional sort on size and TPB

This table presents results for conditional double sorts on size and TPB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TPB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TPB and short stocks with low TPB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	TPB Quintiles					TPB Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.67 [2.89]	0.96 [4.02]	0.83 [4.06]	1.28 [3.46]	0.99 [4.06]	0.32 [2.28]	0.25 [1.76]	0.30 [2.12]	0.23 [1.65]	0.26 [1.81]	0.20 [1.43]
	(2)	0.78 [3.71]	0.79 [4.04]	0.76 [3.67]	0.84 [3.84]	0.81 [3.44]	0.03 [0.27]	-0.05 [-0.39]	0.00 [0.01]	0.02 [0.19]	-0.09 [-0.78]	-0.07 [-0.61]
	(3)	0.68 [3.51]	0.65 [3.61]	0.83 [4.29]	0.87 [4.12]	0.86 [3.81]	0.19 [1.49]	0.07 [0.59]	0.16 [1.37]	0.18 [1.47]	0.05 [0.48]	0.07 [0.60]
	(4)	0.62 [3.34]	0.69 [3.88]	0.71 [3.76]	0.68 [3.54]	0.77 [3.42]	0.15 [1.13]	-0.01 [-0.08]	0.10 [0.83]	0.13 [1.09]	0.02 [0.20]	0.05 [0.43]
	(5)	0.49 [3.16]	0.48 [2.98]	0.54 [3.13]	0.60 [3.31]	0.74 [3.62]	0.25 [1.72]	0.07 [0.51]	0.21 [1.66]	0.21 [1.63]	0.29 [2.31]	0.28 [2.17]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	TPB Quintiles					TPB Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	85	86	86	86	86	6	7	7	8	8	
	(2)	44	44	44	44	44	19	20	20	20	20	
	(3)	39	39	39	39	39	44	44	45	45	44	
	(4)	39	39	39	39	39	119	121	119	119	117	
(5)	43	43	43	43	43	922	1261	1221	1446	1046		

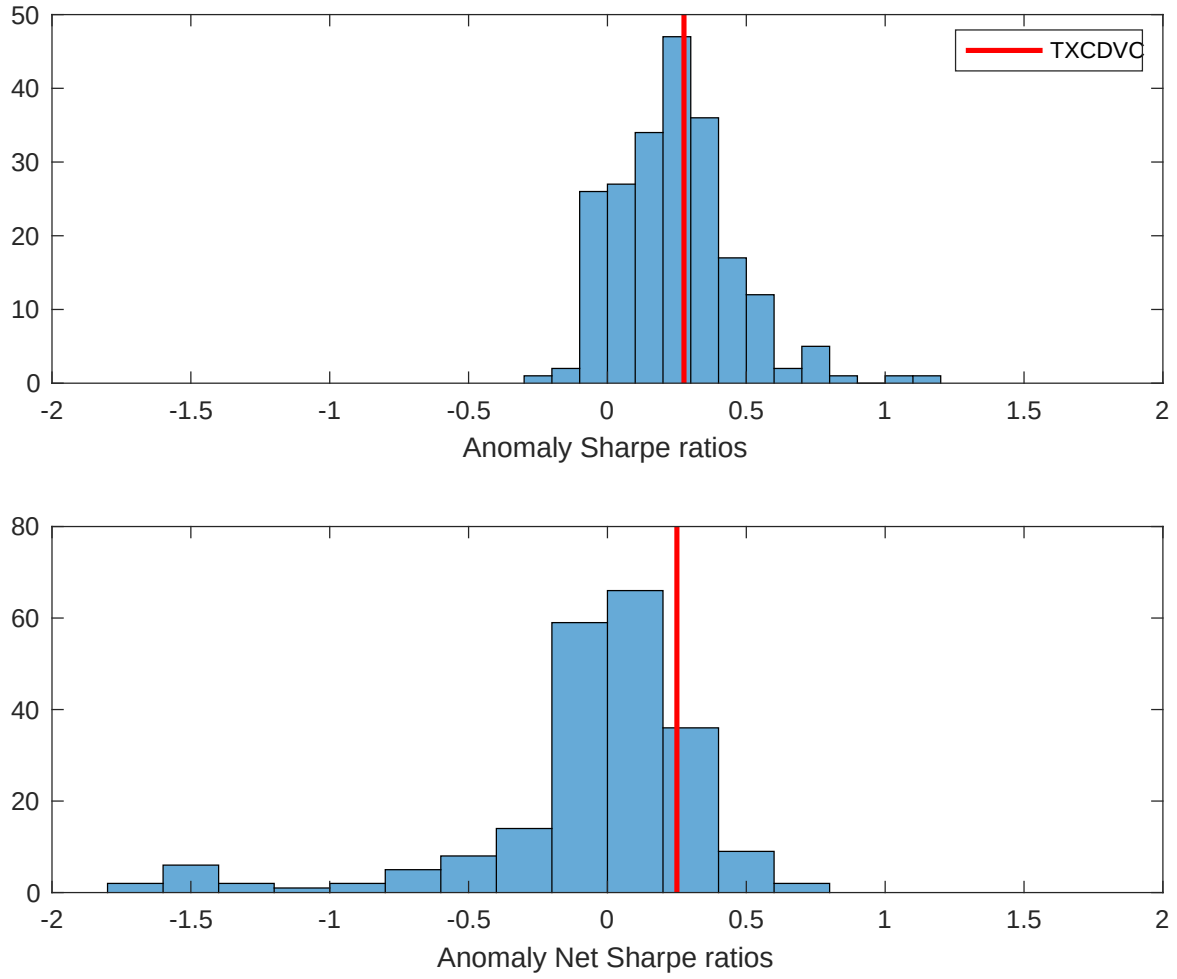


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TPB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

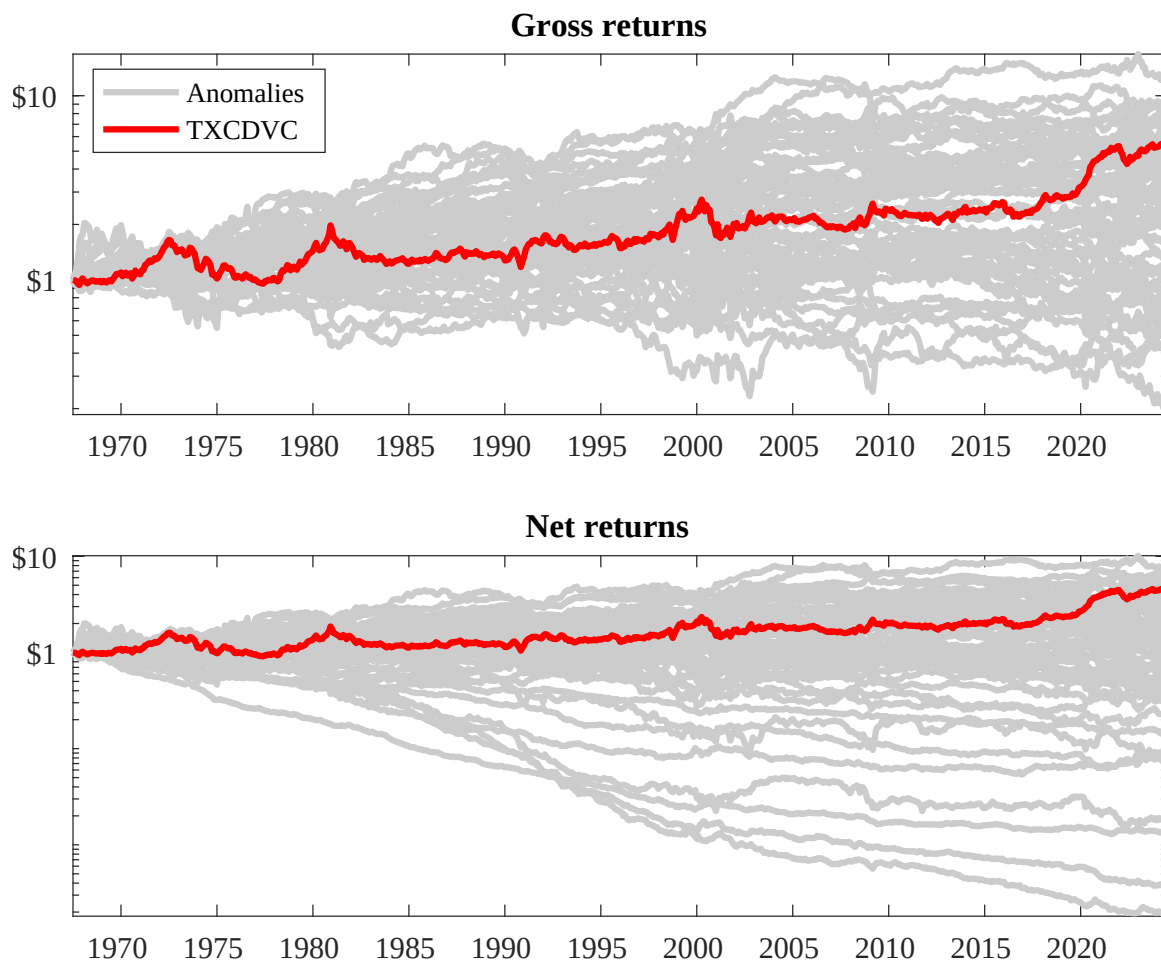
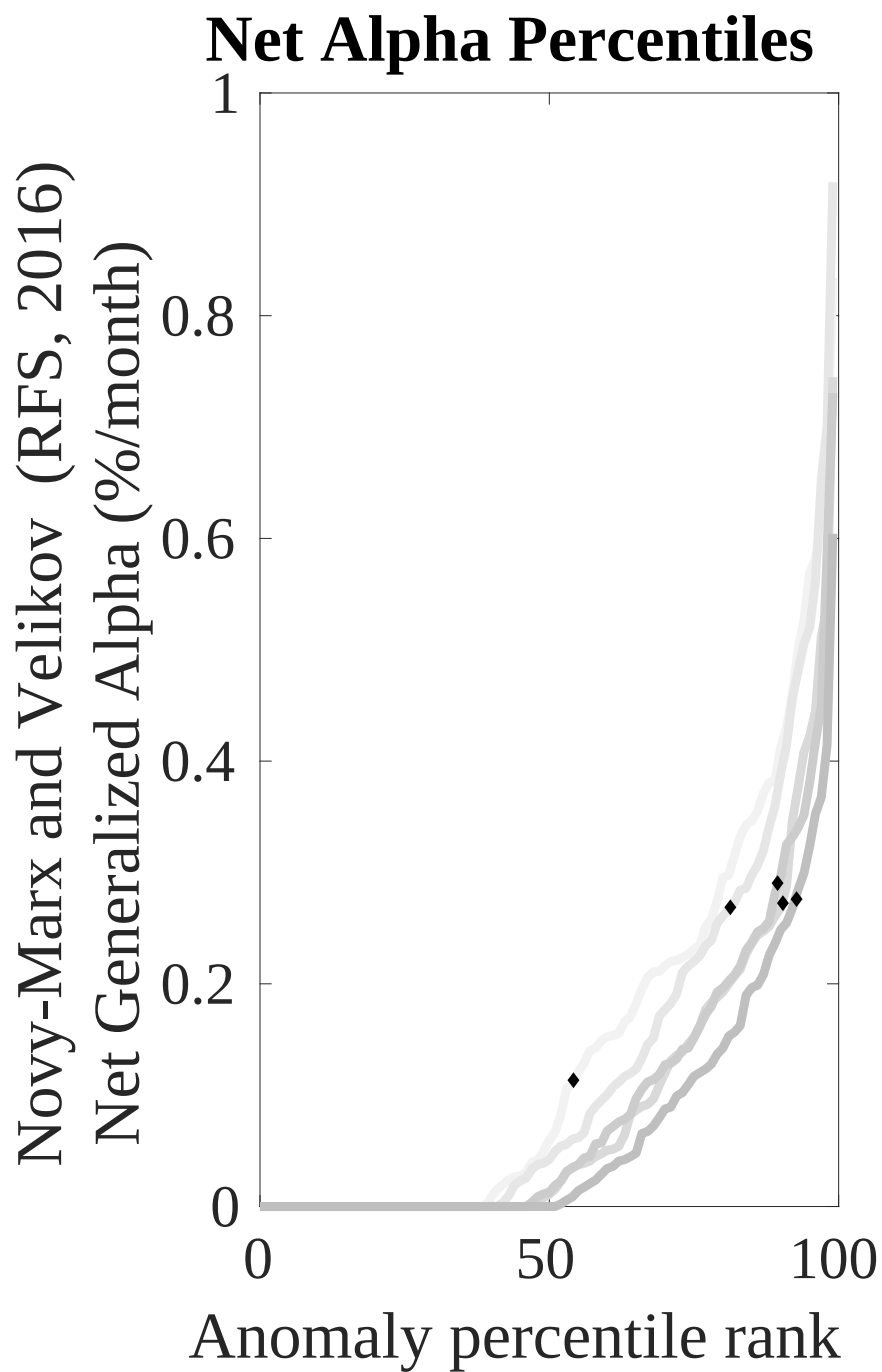
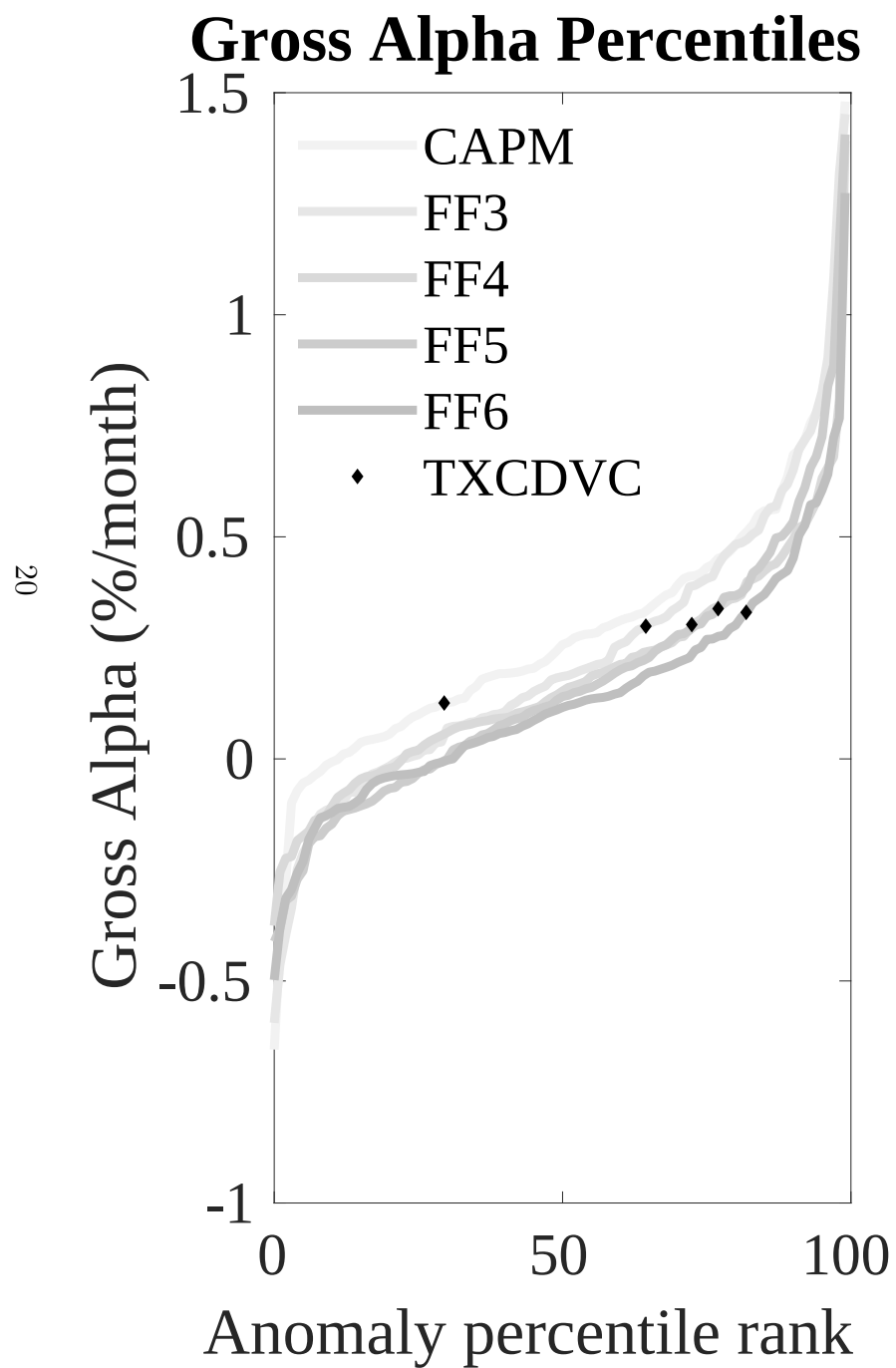


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TPB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



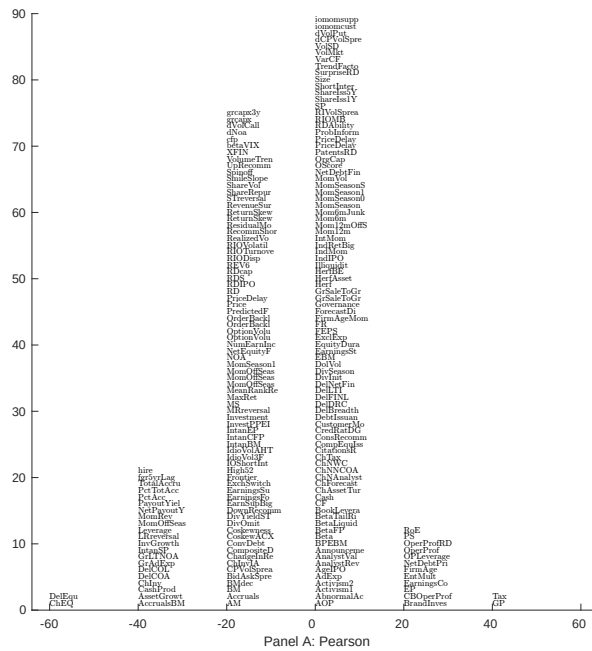


Figure 5: Distribution of correlations.

Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

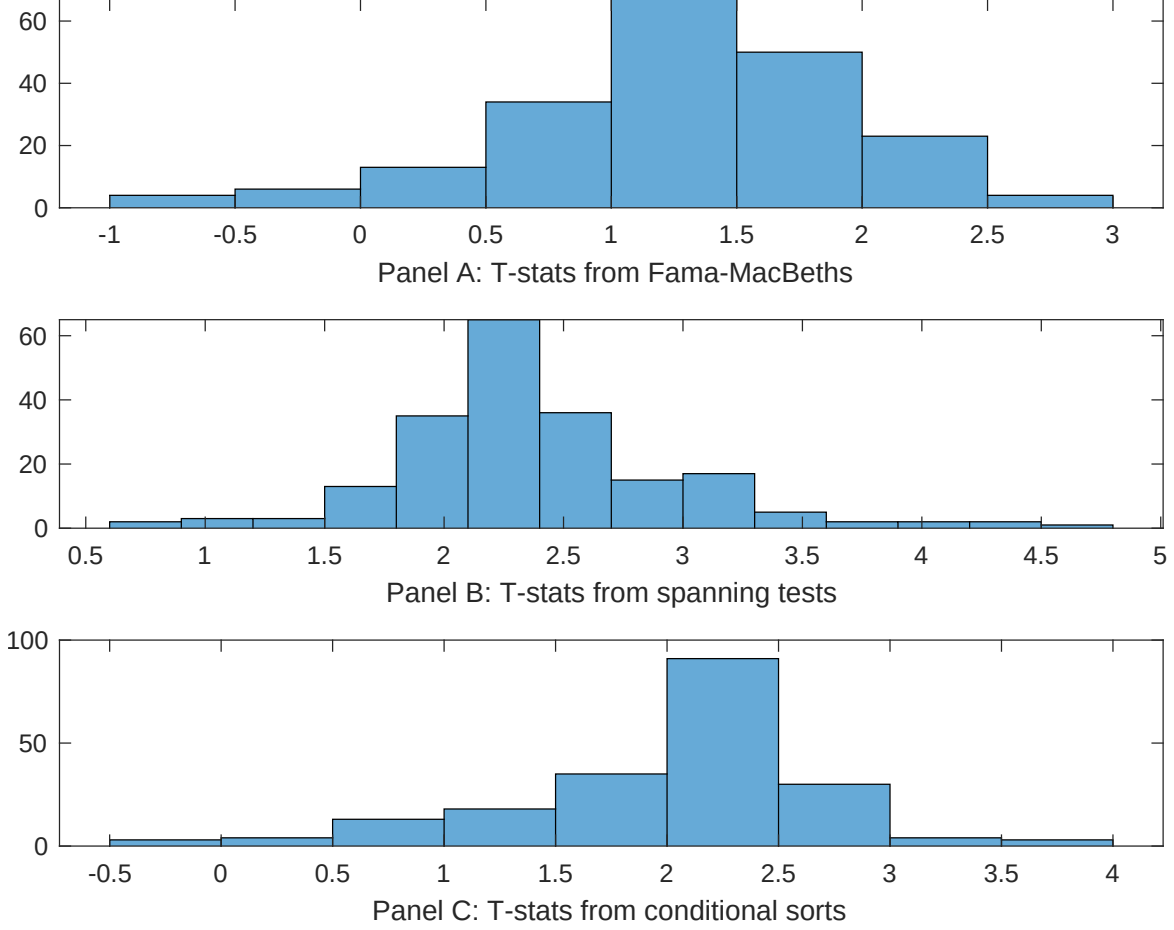


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of TPB conditioning on each of the 201 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TPB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TPB}TPB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 201 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TPB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 201 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 201 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TPB. Stocks are finally grouped into five TPB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TPB trading strategies conditioned on each of the 201 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on TPB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TPB}TPB_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Growth in book equity, Change in equity to assets, Long-term EPS forecast, Payout Yield, Asset growth, Change in current operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.18 [6.61]	0.12 [6.50]	0.13 [7.12]	0.12 [6.21]	0.12 [6.85]	0.12 [6.53]	0.14 [6.76]
TPB	0.29 [2.10]	0.32 [2.27]	0.14 [1.41]	0.27 [2.14]	0.28 [1.97]	0.26 [1.85]	0.21 [1.88]
Anomaly 1	0.59 [2.97]						0.15 [1.08]
Anomaly 2		0.16 [2.50]					-0.29 [-0.53]
Anomaly 3			0.95 [1.26]				0.40 [0.50]
Anomaly 4				0.20 [0.45]			-0.79 [-0.12]
Anomaly 5					0.84 [5.88]		0.38 [2.77]
Anomaly 6						0.13 [3.30]	0.92 [0.17]
# months	684	684	504	668	684	684	499
$\bar{R}^2(\%)$	1	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the TPB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TPB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Growth in book equity, Change in equity to assets, Long-term EPS forecast, Payout Yield, Asset growth, Change in current operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.36 [3.37]	0.34 [3.14]	0.28 [2.17]	0.38 [3.49]	0.34 [3.08]	0.34 [3.03]	0.35 [2.84]
Anomaly 1	-47.18 [-7.90]						-24.25 [-2.17]
Anomaly 2		-42.69 [-7.87]					-9.22 [-0.80]
Anomaly 3			-35.05 [-8.12]				-25.23 [-5.32]
Anomaly 4				-30.71 [-7.91]			-10.45 [-1.93]
Anomaly 5					-40.99 [-6.28]		-7.83 [-0.87]
Anomaly 6						-9.43 [-1.71]	6.80 [1.07]
mkt	11.40 [4.47]	13.70 [5.38]	-1.19 [-0.36]	7.70 [2.94]	12.92 [5.00]	12.99 [4.89]	-1.25 [-0.38]
smb	21.72 [5.79]	20.75 [5.54]	8.56 [1.82]	17.37 [4.64]	23.82 [6.20]	18.47 [4.53]	10.98 [2.26]
hml	-22.05 [-4.41]	-21.81 [-4.35]	-3.50 [-0.58]	-12.25 [-2.31]	-25.68 [-5.09]	-23.66 [-4.22]	-4.81 [-0.76]
rmw	20.46 [4.10]	18.64 [3.72]	24.70 [4.17]	20.82 [4.19]	23.41 [4.62]	21.60 [4.12]	22.67 [3.86]
cma	-6.64 [-0.70]	-9.65 [-1.05]	-32.35 [-3.57]	-39.32 [-5.19]	-4.10 [-0.38]	-48.26 [-5.87]	2.78 [0.23]
umd	2.27 [0.89]	0.45 [0.18]	2.81 [0.97]	-3.71 [-1.43]	0.32 [0.13]	0.95 [0.36]	1.40 [0.47]
# months	684	684	504	680	684	684	500
$\bar{R}^2(\%)$	43	43	41	43	41	38	46

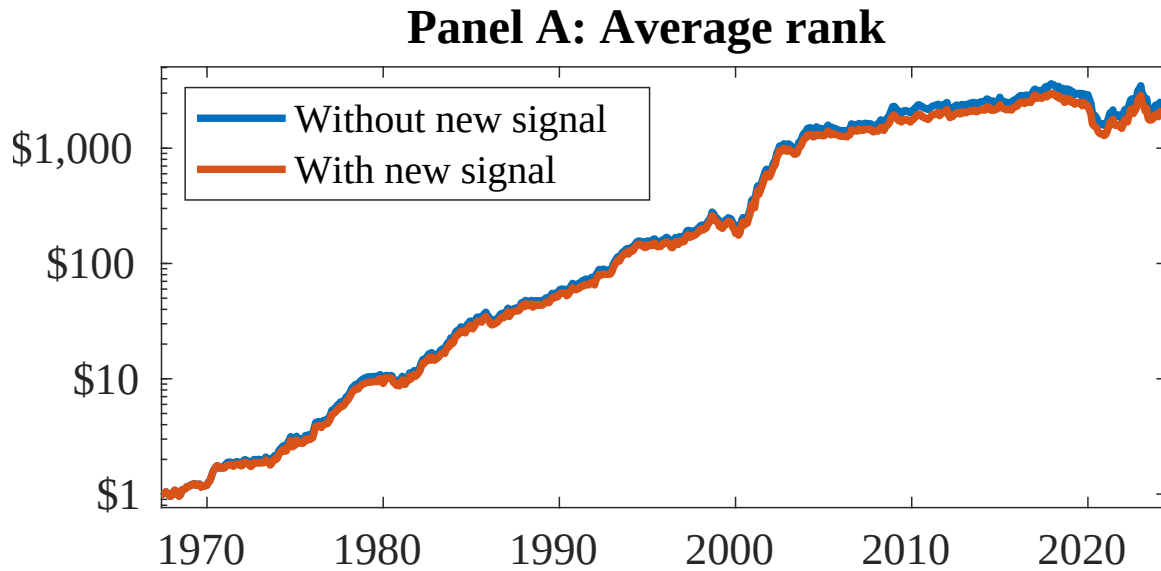


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as TPB. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Bansal, R. and Yaron, A. (2004a). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Bansal, R. and Yaron, A. (2004b). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Lettau, M. and Ludvigson, S. (2001). Consumption, aggregate wealth, and expected stock returns. *Journal of Finance*, 56(3):815–849.
- Lettau, M., Ludvigson, S. C., and Wachter, J. A. (2008). The declining equity premium: What role does macroeconomic risk play? *Review of Financial Studies*, 21(4):1653–1687.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Parker, J. A. and Julliard, C. (2005). Consumption risk and the cross section of expected returns. *Journal of Political Economy*, 113(1):185–222.
- Yogo, M. (2006). A consumption-based explanation of expected stock returns. *Journal of Finance*, 61(2):539–580.